EE4-25
EE9-CS1-2
EE9-FPN2-06

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

Friday, 14 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : E.C. Kerrigan

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1. Consider the feedback system in Figure 1. Here, $G(s)$ is a nominal system model and $K(s)$ is a controller. The transfer matrices $\Delta_a(s)$ and $\Delta_m(s)$ denote stable uncertainties.

The robust stability specification is to design a controller $K(s)$ such that the feedback loop is internally stable when:

- $\Delta_a = 0$ and $\|\Delta_m(j\omega)\| \leq |w_m(j\omega)|, \forall\omega$, and,
- $\Delta_m = 0$ and $\|\Delta_a(j\omega)\| \leq |w_a(j\omega)|, \forall\omega$,

where $w_a(s)$ and $w_m(s)$ are appropriate weighting functions.

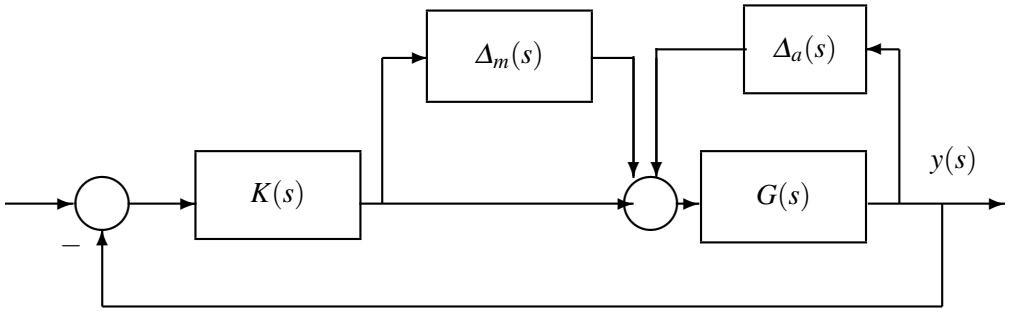


Figure 1

a) State the small gain theorem concerning the internal stability of a feedback loop with a forward transfer matrix Δ and a feedback transfer matrix H. [4]

b) Use the small gain theorem to derive $\mathscr{H}_\infty$–norm bounds, in terms of $G(s)$, $K(s)$, $w_a(s)$ and $w_m(s)$ that are sufficient to achieve the specifications. [4]

c) Define appropriate cost, external, control and measured signals and draw a block diagram, showing these signals, the nominal system model, the controller, and weighting functions. [4]

d) Derive and draw a generalised regulator problem that captures these sufficient conditions. [4]

e) Comment on the expected performance of the resulting design. [4]

2. a) Consider a state–variable model described by the dynamics

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad y(t) = Cx(t) + Du(t),$$

and let $H(s) = D + C(sI - A)^{-1}B$ denote the corresponding transfer matrix. Let $P = P^T$ and

$$M = \begin{bmatrix} A^T P + PA + C^T C & PB + C^T D \\ B^T P + D^T C & D^T D - \gamma^2 I \end{bmatrix}.$$

i) Derive sufficient conditions in the form of LMIs involving P for the stability of A. [4]

ii) Derive sufficient conditions in the form of LMIs involving P and M for

$$\|H\|_\infty < \gamma.$$

[4]

iii) Derive a dual set of conditions for $H(s)$ stable and $\|H\|_\infty < \gamma$. [4]

b) Consider the output injection problem in Figure 2. Let $T_{ed}(s)$ denote the transfer matrix from d to e. An internally stabilizing gain matrix L is to be designed such that, for given $\gamma > 0$, $\|T_{ed}\|_\infty < \gamma$.

i) Derive a state space realization for $T_{ed}(s)$. [4]

ii) Derive sufficient conditions for the existence of a feasible L. Your conditions should be in the form of the existence of solutions to linear matrix inequalities. [4]

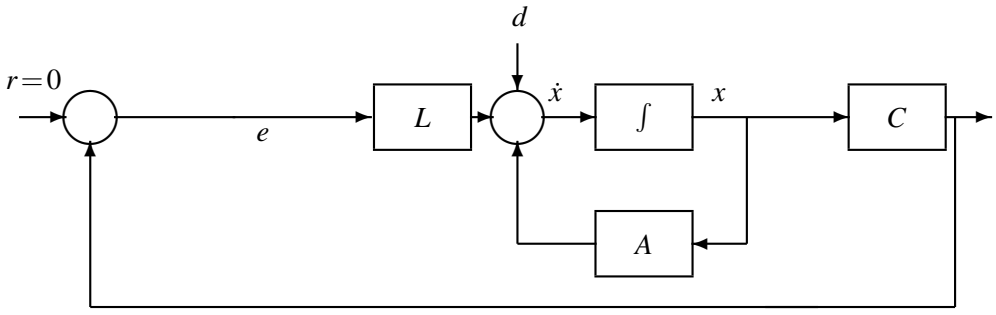


Figure 2

3. Consider a state–variable model described by the dynamics

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t). \end{aligned}$$

a) Suppose that

$$G(s) \stackrel{s}{=} \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] = \left[\begin{array}{ccc|cc} -1 & 0 & 3 & 1 & 0 \\ 0 & 2 & 4 & 0 & 2 \\ 0 & 0 & 5 & 0 & 0 \\ \hline 1 & 0 & 2 & 0 & 0 \\ 0 & 3 & 4 & 0 & 0 \end{array}\right].$$

i) Find the uncontrollable and/or unobservable modes and determine whether the realisation is detectable and stabilisable. [4]

ii) Obtain a minimum realisation of $G(s)$ and evaluate $G(s)$. [4]

iii) Draw a diagram illustrating the nonminimal/minimal subsystems of the state-space realisation. [4]

b) i) Let $A \in \mathscr{R}^{n\times n}$. Suppose there exists a matrix $Q = Q' \succ 0$ such that

$$A'Q + QA \prec 0.$$

Prove that A is stable. [4]

ii) Let σ be a real number. Using Part (bi) above or otherwise, derive sufficient LMI conditions to ensure that the eigenvalues of A are to the left of the line $s = \sigma$ in the complex plane.

[4]

4. Consider the $\mathscr{H}_\infty$ filter for estimating Cx shown in Figure 4.

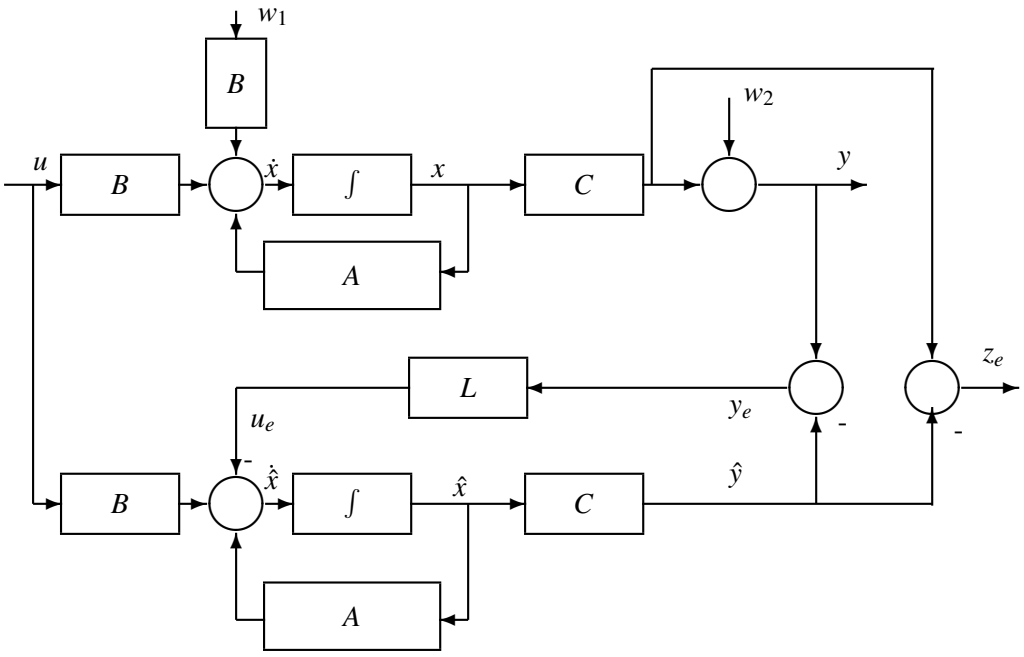


Figure 4

Let $T_{z_e w}(s)$ denote the transfer matrix from $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ to z_e. A stabilizing filter gain matrix L is to be designed such that, for $\gamma > 0$, $\|T_{z_e w}\|_\infty < \gamma$.

a) Derive the generalized regulator system for the filter. Take the state to be $x_e(t) = x(t) - \hat{x}(t)$, the external signal to be $w(s)$, the control input to be $u_e(s)$, the measured output to be $y_e(s)$ and the cost signal to be $z_e(s)$. [4]

b) State duality principles that can transform the filtering problem above into a state feedback problem. [4]

c) Hence derive the generalized regulator system for the dual problem. [4]

d) Suggest a cost function that captures the design requirement for Part (c). Derive an expression for the cost function that allows you to find the solution of the problem. [4]

e) Hence, derive sufficient conditions for the solution of the problem. Your conditions should be in the form of the existence of a certain solution to an algebraic Riccati equation. It should also include an expression for L and an expression for the worst-case disturbance w. [4]